

# San (The) Cao

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## Education

2018–2022 **Vietnam National University Ho Chi Minh City - University of Science**  
Bachelor's Degree in Computer Science  
*Published 2 research papers in AI (SRGAN & Video Classification) at NAFOSTED NICS and MediaEval 2020.*

## Work Experience

- May 2022 – **Software Engineer (Distributed Systems), Zalo, HCMC**  
Present
- Built automated cross-DC replication for **1,000+ in-house DB clusters**, achieving **99.9% consistency** and enabling warm-standby DR.
  - Designed metrics ingestion pipeline at **1M samples/sec** with **10s anomaly detection**; architected Thanos long-term store, cutting query latency 40% and storage cost **3.5x**.
  - **Encryption**: added in-memory encryption to Memcached, securing PII for 70M+ MAU with <2% latency impact.
  - **Backup & Restore**: engineered daily **10TB+** backup flow with automated recoverability tests and snapshot drift metrics vs prod/previous.
  - **Hermetic Build**: created reproducible build pipeline from source to binary, deployed to **2,000+ DB nodes** with zero-drift releases.
  - **Stack**: C++, Java, Linux, Prometheus, Thanos, Grafana, Apache Thrift, Docker, Kubernetes.
- May 2021 – **Software Engineer Intern, VNG Corporation, HCMC**  
Nov 2021
- Developed new features and maintained a card-games called Truco for Argentinians on server side.
  - Maintained tool used for analysing user behavior based on given metric log.
  - Technologies: Java, Memcached, Python, Hadoop, Spark, MySQL.

## Skills

- **Languages**: C/C++, Python, Java, Go, SQL.
- **Technologies**: Linux, Docker, Prometheus, Thanos, Apache Thrift, Kafka.
- **Core Technologies**: Distributed Systems, Data Structures/Algorithms, System Reliability Engineering.

## Projects

- 2020 – 2021 **Free Contest, Backend Developer & Problem Setter**  
Designed and developed API servers (TypeScript, ExpressJS, MySQL) and prepared competitive programming problems for weekly contests.
- 2018 **cowboycode.tech**  
A website sharing problem solution and code, as well as series of learning tips written or translated in Vietnamese.

## Honours and Awards

- 2021 **Top 10 in AIOT Innovation Works, HCMC**  
2020 **Fourth prize in AI Challenge of AISocha 2020, HCMC**  
2018 **Second Prize in ACM-ICPC Vietnam Southern Provincial Contest**  
2018 **Third Prize in Vietnamese National Olympiad in Informatics**